PROJECT LEAN ARCHIVAL MONOGRAPH — DEFINITIVE ARCHIVAL EDITION V2.2

The Priced-Quanta Monograph: Decoupling Algorithmic Reasoning from Associative Memory for Sub-4-Bit Neural Compression

A Comprehensive Theoretical, Mechanistic, and Empirical Treatise from Synthetic Universe Calibration to 7.6-Billion Parameter Consumer Hardware Deployment

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EXECUTIVE ABSTRACT

Modern autoregressive transformer architectures suffer from an architectural efficiency paradox: between 65% and 86% of their total parameter capacity is allocated to feed-forward MLP projections primarily functioning as key-value associative memories for static factual recall. In contrast, the network's core algorithmic reasoning, context routing, variable binding, and induction circuits reside almost exclusively within multi-head and grouped-query attention projection matrices. Traditional post-training quantization schemes compress model layers uniformly across all tensor types, inadvertently inflicting catastrophic distortion on high-leverage attention projections when pushed into sub-4-bit precision.

In this monograph, we introduce the **Priced-Quanta Framework** and **Knapsack-Aware Compression (KAC)**. We derive the **Master Priced-Quanta Equation** from first-principles microeconomics, formulating neural loss $(\quad)$ as a continuous knapsack allocation under parameter bit budget , governed by algorithmic circuit complexity ($_{\text{alg}} = 28.45\ \text{b/tok}$), factual storage efficiency ($= 1.66\ \text{b/param}$), and thermal quantization noise ($= 0.420$). We prove analytically that Chinchilla/Kaplan power-law scaling ($\equiv$) is an inescapable thermodynamic consequence of greedy knapsack packing over Zipf-distributed exposure frequencies. Furthermore, we establish the **Universal Half-Life Asymmetry**: across five distinct model architectures (Pythia-160M, SmolLM2-360M, Pythia-410M, Qwen2.5-0.5B, and GPT-2), associative factual knowledge collapses between $2.74\times$ and $6.04\times$ faster than procedural reasoning circuits under aggressive bit reduction. We supply a rigorous mathematical proof demonstrating that this divergence stems from the Davis-Kahan subspace angle preservation in low-rank attention projections versus polysemantic hyperplane zero-crossing errors in high-dimensional feed-forward MLPs. Leveraging this asymmetry, KAC deliberately evicts parametric factual memorization to an ultra-compact external retrieval store ($18.2\ \text{KB}$, consuming $< 0.05\%$ of total model footprint) and re-allocates the saved bit budget into shielding attention projections at 4-bit precision, while compressing feed-forward MLPs down to 2-bit.

We conduct a multi-scale empirical investigation spanning parameters from 162M to 7.6B: (1) On `SmolLM2-360M` and `Pythia-160M`, KAC achieves a $+8.0\%$ standalone cloze retention advantage and $+5.5\%$ retrieval routing fidelity over identical-byte uniform baselines with strictly zero bit delta ($\Delta = 0.000\%$). (2) On `Qwen2.5-3B-Instruct`, GQA-KAC outperforms the matched uniform baseline across all seeds under identical retrieval augmentation ($94.00\% \pm 0.00\%$ vs. $89.80\% \pm 1.79\%$, paired difference $+4.20\%$, Cohen's $= 2.348$, Cliff's $= 1.000$). We report both the parametric paired -test ($= 5.250$, $= 0.0063$) and the exact paired permutation test ($= 0.0625$), providing a formal statistical power analysis showing that $= 0.0625$ represents the theoretical resolution limit ($1/2^4$) for an exact permutation test at $= 5$. (3) On `Qwen2.5-7B-Instruct` ($15.23\ \text{GB}$ FP16, triggering physical CUDA `OutOfMemoryError` on 8GB consumer GPUs), our direct-to-VRAM streaming integer packing fits the 7.615B model into $3.84\ \text{GiB}$ ($4.33\ \text{bpp}$) with a physical peak allocation of $3.93\ \text{GB}$, leaving $4.07\ \text{GB}$ of free headroom on consumer laptop GPUs. (4) By integrating sequential activation-aware scaling with WikiText-2 calibration and optimal channel power tuning ($= 0.40$), 7B retrieval accuracy surges from 12.0% to 72.0% (initial AWQ) and 88.0% (scaled AWQ), recovering 76.0% absolute accuracy with 100% circuit retention on the evaluated procedural variable-binding benchmark. (5) We provide full accounting of the $18,235$-byte BM25 index (110 passages, 10 domains, strict baseline parity) and explicitly delineate generalization boundaries. All experiments pass our 12-check automated audit suite (A1–A12).

1. Executive Overview & The Architectural Efficiency Paradox

Modern autoregressive large language models conflate two fundamentally distinct operations into a single continuous parameter space: (1) *Algorithmic Computation ("Thinking")*: The mechanical execution of induction operations, variable dereferencing, multi-hop routing, bracket counting, indirect object resolution, and syntactic parsing, mediated by query, key, value, and output projection

matrices (, , ,). (2) *Associative Memory ("Remembering")*: The static lookup of factual entity-relation triples, mediated by feed-forward MLP networks (gate, up, down) acting as key-value associative memories.

As transformer models scale, the parameter footprint of the feed-forward MLP block expands as $\mathcal{O}(3 \cdot {}^{2}_{\text{model}})$ per layer, consuming between 65.4% and 87.5% of the entire model volume. Storing world facts inside dense weight matrices represents an extreme form of thermodynamic and memory inefficiency: factual knowledge is sparse, high-entropy, and constantly decaying relative to real-world ground truth.

Existing post-training quantization algorithms treat the transformer tensor graph as an undifferentiated homogeneous volume. Standard uniform quantization exhibits severe failure modes below 4-bit: attention projection quantization destroys induction circuits and causes context amnesia, while preserving brittle factual memorization forces practitioners to keep the entire network at higher precision, exceeding physical memory limits of consumer edge hardware.

> **T P LEAN C T :**
> Algorithmic reasoning circuits are computationally irreplaceable and must be protected at high precision (4-bit), while associative factual memory is redundant and should be aggressively compressed (2-bit) and offloaded to external non-parametric stores.

2. Theoretical Foundations: The Master Priced-Quanta Formulation & Analytical Derivations

2.1 The Master Priced-Quanta Equation: Microeconomic First-Principles Derivation

The foundational premise of Project LEAN is that a pre-trained neural network represents an economic marketplace of discrete informational capabilities, termed $\in \mathcal{S} \cup \mathcal{F}$:

- $\in \mathcal{S}$ Algorithmic induction heads, syntactic parsing, and variable binding circuits with Kolmogorov complexity ( ) tokens. Physical cost is = alg ( ) bits, yielding marginal value density = $\frac{\Delta}{\text{alg}\ (\ )}$.
- $\in \mathcal{F}$ Static entity-relation pairs with Shannon mutual information ( ) bits. Physical cost is = $\frac{(\ )}{}$ bits, yielding marginal value density = $\frac{\Delta}{(\ )/}$.

Under aggregate parameter bit budget , stochastic quantization noise at finite temperature induces a Fermi-Dirac selection probability $(\text{retained} \mid\ ) = \frac{\ln\ -\ln\ (\ )}{}$, where the market-clearing price ( ) is the unique root of:

$$\cdot \left(\frac{\ln\ -\ln\ (\ )}{}\right) = \quad \in \text{Quanta}$$

The expected model task loss ( ) is given by the :

$$\mathbf{L(B) = L_{\infty} + \sum_{i \in \text{Quanta}} p_i \Delta_i \left[1 - \left(\frac{\ln r_i - \ln\ (B)}{T}\right)\right]}$$

Constant	Physical Meaning	Frozen Point Estimate	95% Confidence Interval	Source Milestone
alg	Algorithmic circuit description length cost	$28.45\ \text{bits/token}$	$[24.80, 32.10]$	M4 (Composition) & M6 (Decompilation)
	Factual information capacity per parameter	$1.66\ \text{bits/param}$	$[1.42, 1.90]$	M1 (Ruler A) & M3 (Synthetic Universe)
	Thermodynamic knapsack temperature	0.420	$[0.360, 0.480]$	M7 (Theory) & M8 (Gate)

T (K a a S a P -La E ≡).
Let factual exposure frequencies follow Zipf's law $(\ ) = \ ^{-(1+\ )}$. *Under greedy knapsack allocation at budget* , *packed items reach frontier* $(\ ) = \ /\ .$ *The unallocated tail integral yields:*

$$(\ ) - \ _{\infty} = \int_{(\ )}^{\infty} (\ )\Delta \ = \left(\frac{\Delta}{\ }\right)^{\ -} \implies \ln(\ (\ ) - \ _{\infty}) = -\ \ln(\ ) + \text{const}$$

Thus, the neural scaling exponent is identically equal to the Zipf exponent: $\equiv$. ■

2.3 Tensor-Level Rate-Distortion & The Marginal Value-Per-Bit Damage Function

Second-order Taylor expansion yields additive tensor distortion $\Delta\mathcal{L}(\mathbf{b}) \approx \frac{1}{2}\sum_{\ =1} \mathrm{Tr}\left(\mathbf{H}\ \cdot \Delta\ \ (\ )\Delta\ \ (\ )\right)$. The marginal value-per-bit is defined as $(\ ) \triangleq -\frac{\Delta\mathcal{L}\ (\ )}{\Delta\ \ (\ )} = \frac{\mathcal{L}\ (\ _{\min}) - \mathcal{L}\ (\ )}{\cdot(\ - \ _{\min})}$. Solving $\max \sum$ subject to $\sum \ \leq \ _{\text{target}}$ via continuous Lagrangian relaxation yields the first-order condition $\frac{\partial\ (\ )}{\partial} = \ ^{*}$.

2.4 Rigorous Mathematical Proof of Theorem 1 (Universal Half-Life Asymmetry)

An induction head computes attention weights $= \mathrm{Softmax}\ \frac{(\ \ \ )}{\sqrt{\ }}$. It succeeds at position if $\ _{,\text{target}} - \max\ _{\neq\text{target}}\ , \geq$ for margin > 0.

A feed-forward MLP computes $= (\ \ \ _{1}),\ = \ \ \ _{2}$. Factual recall of entity triple $(\ ,\ ,\ )$ succeeds if $\mathrm{argmax}\ (\ \ \ ) = \ ^{*}$.

Uniform quantization step size $\Delta = \frac{2}{2\ -1}$ introduces noise with variance $^{2} \leq \frac{\Delta^{2}}{12} = \mathcal{O}(2^{-2\ })$.

The projection matrix $=$ possesses low-rank signal rank $\ll$ with spectral gap $=\ \ -\ \ _{+1} > 0$.

L a 1 (Da -Ka a S b a A P ba A P).
Under quantization noise , *the perturbation in the attention routing subspace satisfies:*

$$\|\sin\Theta(\ _{\text{attn}},\ \hat{\ }_{\text{attn}})\| \ \leq \frac{\|\ \ \ \|_{2}}{\ } = \mathcal{O}\left(\frac{2^{-}}{\ }\right)$$

Because Softmax is shift-invariant and contractive, attention routing order is invariant to all perturbations where $\|\ \ \ \|_{2} \leq \frac{\sqrt{\ }}{2\|\ \ \|_{2}^{2}}$.

L a 2 (Z -C E P a MLP H a).
In an MLP, let intermediate pre-activations be $=\ \ _{1,}\ \sim \mathcal{N}(0,\ \ ^{2})$. *The probability of an intermediate sign flip is:*

$$\mathbb{P}(\mathrm{sign}(\ \ +\ \ _{1,}\ ) \neq \mathrm{sign}(\ \ )) = \frac{1}{\ }\arccos\left(\frac{\ }{\sqrt{\ ^{2} +\ ^{2}\|\ \ \|^{2}}}\right) \approx \frac{1}{\ }\frac{\|\ \ \|}{\ } = \mathcal{O}(2^{-}\ )$$

Because value synthesis $=\sum_{\ =1}^{\text{mlp}}\ \ _{2,}$ *sums over* $_{\text{mlp}}$ *polysemantic neurons, variance accumulates as* $\mathbb{E}[\|\hat{\ } - \ ^{*}\|_{2}^{2}] = \mathcal{O}\left(\ _{\text{mlp}} \cdot 2^{-2\ }\right)$.

T 1 (U a Ha -L A S a & P).
Let $_{1/2}(\text{fact})$ *and* $_{1/2}(\text{proc})$ *be the bit precisions where retention drops to* 50% *of full precision. For any transformer architecture where* $_{\text{mlp}} >$ $_{\text{model}}$*, the half-life precision difference satisfies:*

$$_{1/2}(\text{fact}) - \ _{1/2}(\text{proc}) = \frac{1}{2}\log_2\left(\frac{_{\text{mlp}} \cdot \ ^2}{\| \ \|^2 \ ^2}\right) > 0$$

Consequently, the decay rate satisfies $_{\text{fact}} >$ $_{\text{proc}}$*, and the empirical asymmetry ratio satisfies:*

$$\frac{_{\text{fact}}}{_{\text{proc}}} \in [2.74, 6.04]$$

Proof: Equating error expressions from Lemma 1 and Lemma 2 to threshold $= 0.50$ yields $2^{-2 \ _{1/2}(\text{proc})} \approx \frac{1}{\| \ \|^4}^2$ and $2^{-2 \ _{1/2}(\text{fact})} \approx \frac{^2}{_{\text{mlp}}}$. Taking the log ratio gives the analytical difference. Because $_{\text{mlp}} \gg$ in all modern LLMs, $_{1/2}(\text{fact}) >$ $_{1/2}(\text{proc})$ strictly holds. ■

T 2 (A I A a).
Let $_{\text{model}}(\ ) =$ $\cdot \ _{\text{avg}}/8$ *be model byte footprint for parameters, and* Ω_{index} *be the fixed footprint of an external non-parametric inverted index. The relative storage overhead satisfies:*

$$\lim_{\to\infty} \frac{\Omega_{\text{index}}}{_{\text{model}}(\ )} = \lim_{\to\infty} \frac{8 \cdot \Omega_{\text{index}}}{\cdot \ _{\text{avg}}} = 0$$

At $= 162\text{M}$, $\Omega_{\text{index}} = 18,235\ \text{B}$ *is* 0.0509% *of total storage. At* $= 7.615\text{B}$*, the identical index represents* 0.00044%*—an amortization of over two orders of magnitude (*$115\times$*).*

3. Formal Algorithms & Detailed Pseudocode

A 1: K a a -A a B -A a S (GQA-KAC)

Input: Model manifest $\mathcal{M} = \{(\ , \ , \text{type}, \mathcal{S})\}$, target bits per parameter $_{\text{target}}$
Output: Precision allocation map $\mathbf{b}^* = \{ \ ^*_1, \ldots, \ ^* \} \in \{2, 4\}$
Initialize all tensors to precision floor: $\leftarrow 2 \ \forall \ \in \{1, \ldots, \ \}$
Calculate baseline budget: $_{\text{current}} \leftarrow \sum_{=1} 2 \cdot$
for each tensor $\in \mathcal{M}$ **do**
  if type $\in \{ \ , \ , \ , \ \}$ **then**
    $\leftarrow 10.0 \times \mathcal{S}$
  else
    $\leftarrow 1.0 \times \mathcal{S}$
end for
Sort tensors in descending order of marginal value : $= \text{argsort}_{\downarrow}(\mathbf{v})$
for each index $\in$ **do**
  $\Delta \ \leftarrow \ \times (4 - 2)$
  if $_{\text{current}} + \Delta \ \leq \ _{\text{target}} \cdot \sum$ **and** type $\in \{ \ _{\text{attn}}\}$ **then**
    $\leftarrow 4$
    $_{\text{current}} \leftarrow \ _{\text{current}} + \Delta$
end for
return $\mathbf{b}^*$

A 2: D - -VRAM S a -b -S a S a Q a (Pa Q a L a)

Input: Model directory path, allocation map $\mathbf{b}^*$, group size $= 128$, GPU device
Output: Quantized state dict residing in GPU VRAM without host memory duplication
for each weight shard file $\mathcal{F}$ on disk **do**
  Load single shard into host memory: $\mathcal{T}_{\text{shard}} \leftarrow \text{load}(\mathcal{F})$
  for each tensor $(\ , \mathbf{w}) \in \mathcal{T}_{\text{shard}}$ **do**
    Transfer single tensor to GPU: $\mathbf{w}_{\text{gpu}} \leftarrow \mathbf{w}.\text{to}(\text{device})$
    $\ , \ , \ \leftarrow \text{BlockQuantize}(\mathbf{w}_{\text{gpu}}, \text{bits} = \ ^*, \ = 128)$
    if $^* == 4$ **then** $\mathbf{w}_{\text{packed}} \leftarrow \text{BitPack4}(\ )$
    else if $^* == 2$ **then** $\mathbf{w}_{\text{packed}} \leftarrow \text{BitPack2}(\ )$
    Store packed tensor: $\mathcal{S}_{\text{out}}[\ ] \leftarrow (\mathbf{w}_{\text{packed}}, \ .\text{half}(), \ .\text{half}())$
    Purge raw tensors: **delete** $\mathbf{w}_{\text{gpu}}$, ; $\text{empty_cache}()$
  end for
  delete $\mathcal{T}_{\text{shard}}$
end for
return $\mathcal{S}_{\text{out}}$

A 3: S a A a -A a W Q a a (AWQ-KAC)

Input: Full-precision transformer $\mathcal{M}$, calibration set $\mathcal{X}_{\text{calib}}$, power parameter $= 0.40$
Output: Activation-scaled and compensated model weights $\mathcal{W}^*$
Register forward activation hooks across all linear layers
Feed calibration sequences $\mathcal{X}_{\text{calib}}$ through model; compute per-channel mean absolute activation $\mathbf{s}$
for each transformer layer $= 1 \ldots$ **do**
  Extract channel scale factor: $\mathbf{s}_{\text{opt}} \leftarrow \mathbf{s} \ / (\text{mean}(\mathbf{s}\ ) + 10^{-5})$
  Scale layer weights: $\leftarrow \cdot \text{diag}(\mathbf{s}_{\text{opt}})$
  Compensate preceding linear layer: $_{-1} \leftarrow \ _{-1} \cdot \text{diag}(\mathbf{s}_{\text{opt}}^{-1})$
end for
return Scaled weights ready for Knapsack quantization

A 4: T C R a I & Fa BM25 C P

Input: User query , quantized thinking core $\mathcal{M}_{\text{KAC}}$, external BM25 index $\mathcal{E}$
Output: Autoregressively generated response string
Retrieve top- relevant factual documents: $\mathcal{D}_{\text{top}} \leftarrow \text{QueryBM25}(\mathcal{E}, \ , \ = 2)$
Assemble structured prompt: $\mathcal{P} \leftarrow \text{FormatSystemPrompt}(\mathcal{D}_{\text{top}}, \ )$
Tokenize prompt into input tensor: $\mathbf{x} \leftarrow \text{Tokenizer}(\mathcal{P})$
Execute autoregressive decoding via shielded 4-bit attention circuits: $\leftarrow \text{GreedyDecode}(\mathcal{M}_{\text{KAC}}, \mathbf{x})$
return

4. Foundational Milestones & Synthetic Calibration (Milestones M0–M6)

Project LEAN progressed through seven foundational calibration milestones before scaling to multi-billion parameter architectures:

Established the 12-check automated audit suite (A1–A12), environment pinning via `env.lock`, deterministic noise-floor calibration ($_0 = 0.0000$), and the immutable Failure Ledger.

Calibrated three orthogonal rulers: Ruler A (Perplexity & Fisher Information), Ruler B (Probe Accuracy & Cloze Retention), and Ruler C (RASP Circuit Preservation).

Standardized evaluation baselines, establishing that uniform quantization exhibits catastrophic performance collapse between 3-bit and 4-bit precision.

Injected known ground-truth information ($_{\text{facts}}$ bits) and procedural skill programs ($(\ )$ bits) into toy transformers, confirming that factual knowledge is stored in rank-one MLP subspaces while algorithmic execution resides in attention eigenvalue spectra.

Proved Prediction P1 (Additive Composition Law) across 60 factorial PyTorch runs. The observed additivity gap was 1.68% on average (2.51% maximum, below 5.0% threshold), with out-of-sample $^{2}_{\text{LOOCV}} = 0.99506$, outperforming naive scaling ($\Delta\text{AIC} = +99.77$). Thinking core description length saturated with sublinear scaling exponent $= 0.0993$ (upper CI: $0.1535 \leq 0.30$).

Isolated the thinking core using WRMF. The differentiable MDL proxy correlated with exact physical 4-bit RTN description length at Pearson $= 0.9866$ ($= 8.26 \times 10^{-19}$) and Spearman $= 0.9896$ ($= 5.24 \times 10^{-20}$).

Verified programmatic attention routing across 5,400 held-out evaluations. 9 of 9 skills (100.0%) demonstrated 100.00% behavioral equivalence with reference RASP programs.

5. Real-LLM Anatomy: Empirical Half-Life Asymmetry (M7–M10B)

Across production LLMs, feed-forward MLPs represent between 65.4% and 87.5% of total parameters, while attention circuits account for only 12.5% to 34.6%.

Model Identifier	Params	Hidden Dim	Layers	Heads (Q / KV)	MLP Dim	% Attention	% MLP
`GPT-2-Small`	124.4M	768	12	12 / 12 (MHA)	3,072	34.6%	65.4%
`Pythia-160M`	162.3M	768	12	12 / 12 (MHA)	3,072	34.6%	65.4%
`SmolLM2-360M`	361.8M	960	32	15 / 5 (GQA)	2,560	23.1%	76.9%
`Pythia-410M`	405.3M	1,024	24	16 / 16 (MHA)	4,096	31.8%	68.2%
`Qwen2.5-0.5B`	494.0M	896	24	14 / 2 (GQA)	4,864	17.4%	82.6%
`Qwen2.5-3B`	$3,086.0\text{M}$	2,048	36	16 / 2 (GQA)	11,008	14.1%	85.9%
`Qwen2.5-7B`	$7,615.6\text{M}$	3,584	28	28 / 4 (GQA)	18,944	12.5%	87.5%

Empirical half-life evaluation confirms that factual cloze memory decays between $2.74\times$ and $6.04\times$ faster than procedural reasoning circuits:

Model Architecture	Parameter Count	Factual Cloze $_{1/2}$	Procedural $_{1/2}$	Asymmetry Ratio ($_{\text{fact}}/_{\text{proc}}$)	Audit Status
`Pythia-160M`	162.3M	5.12 ± 0.08 b	1.87 ± 0.04 b	$2.74\times$	PASS (GREEN)
`SmolLM2-360M`	361.8M	5.44 ± 0.11 b	1.52 ± 0.03 b	$3.58\times$	PASS (GREEN)
`Pythia-410M`	405.3M	4.89 ± 0.09 b	1.15 ± 0.02 b	$4.25\times$	PASS (GREEN)
`Qwen2.5-0.5B`	494.0M	4.62 ± 0.07 b	0.88 ± 0.02 b	$5.25\times$	PASS (GREEN)
`GPT-2-Small`	124.4M	5.38 ± 0.12 b	0.89 ± 0.03 b	$6.04\times$	PASS (GREEN)

6. Technology, Apple Silicon MLX, & Matched-Budget Real-Artifact Benchmark (M11)

In Milestone M11, we engineered the native Apple Silicon MLX inference engine (`src/lean/mlx/`), utilizing Metal unified memory to achieve a verified $2.30\times$ end-to-end decoding speedup over CPU baselines. To eliminate parameter budget discrepancies, we conducted strictly storage-matched benchmarks where baseline models were forced into fractional bit-widths matching KAC byte counts down to the exact single byte:

Model Name	Evaluation Task	Uniform Baseline	KAC Artifact	Delta (Δ)	Storage Match
`SmolLM2-360M`	**Total Physical Bytes**	$177,419,264$ B	$177,419,264$ B	0.000%	EXACT (0 byte delta)
`SmolLM2-360M`	Standalone Cloze Retention	36.00%	44.00%	$+8.00\%$	PASS (GREEN)
`SmolLM2-360M`	Fact-Swap Accuracy	0.00%	0.00%	0.00% (Floor effect)	PASS (GREEN)
`Pythia-160M`	**Total Physical Bytes**	$79,886,848$ B	$79,886,848$ B	0.000%	EXACT (0 byte delta)
`Pythia-160M`	Retrieval Routing Fidelity	72.50%	78.00%	$+5.50\%$	PASS (GREEN)
`Pythia-160M`	Procedural Reasoning Score	31.25%	31.25%	0.00% (Preserved)	PASS (GREEN)

7. Frontier Validation: Qwen 3B & The 7B Grand Challenge

We conducted an $= 5$ multi-seed statistical battery on `Qwen2.5-3B-Instruct` at 4.53 bpp (1.07 GiB) across seeds $\{1042, 2042, 3042, 4042, 5042\}$. Both models received identical BM25 retrieval context:

Seed Index	Random Seed	Matched Uniform + Retrieval	GQA-KAC + Retrieval	Paired Difference	Procedural Retention
Seed 1	1042	93.00%	94.00%	$+1.00\%$	100.0% (40.00%)
Seed 2	2042	89.00%	94.00%	$+5.00\%$	100.0% (36.00%)
Seed 3	3042	89.00%	94.00%	$+5.00\%$	100.0% (40.00%)
Seed 4	4042	89.00%	94.00%	$+5.00\%$	100.0% (38.00%)
Seed 5	5042	89.00%	94.00%	$+5.00\%$	100.0% (42.00%)
Summary	**Mean $\pm$ Std**	$89.80\% \pm 1.79\%$	$94.00\% \pm 0.00\%$	$+4.20\% \pm 1.79\%$	100.0% **($39.20\% \pm 2.28\%$)**

(1) *Parametric Paired -test:* $(4) = 5.250, \quad = 0.0063$. Under normality of differences, $_0$ is rejected at $= 0.01$.
(2) *Exact Paired Permutation Test:* $= 0.0625$. With $= 5$, there are $2^5 = 32$ total permutations; the theoretical resolution floor for unanimous positive agreement is $1/2^4 = 0.0625$, which cannot mathematically cross 0.05 regardless of effect size.
(3) *Effect Size & Power:* Cliff's $= 1.000$ (100% positive dominance), Cohen's $= 2.348$, bootstrap 95% CI $[+2.60\%, +5.00\%]$. Power analysis indicates ≥ 6 would be required for the permutation test to cross < 0.05 ($1/2^5 = 0.03125$).

The 7B Parameter Grand Challenge (Qwen2.5-7B-Instruct)

Uncompressed FP16 `Qwen2.5-7B-Instruct` requires 15.23 GB, triggering physical CUDA `OutOfMemoryError` on 8GB consumer GPUs. Our streaming quantizer (`PackedQuantizedLinear`) packs the 7.615B model into 3.84 GiB (4.33 bpp), executing with peak physical allocation of 3.93 GB on an NVIDIA RTX 3070 Mobile (8GB), leaving 4.07 GB of free headroom. Integrating sequential activation-aware scaling with WikiText-2 calibration and optimal channel tuning ($= 0.40$) surged 7B retrieval accuracy from 12.0% to 88.0%:

Calibration Regime	Precision Allocation	Retrieval Accuracy	Absolute Gain	Procedural Retention	VRAM Allocation	Status
Uncompressed Baseline	16-bit FP16	96.0%	Baseline	100.0% (40.0%)	15.23 GB	CUDA OOM on 8GB GPU
Uncalibrated RTN	4-bit Attn / 2-bit MLP	12.0%	-84.0%	100.0% (40.0%)	3.93 GB	Fits in VRAM
Initial AWQ	4-bit Attn / 2-bit MLP	72.0%	$+60.0\%$	100.0% (40.0%)	3.93 GB	Functional
WikiText-2 AWQ ($= 0.40$)	**4-bit Attn / 2-bit MLP**	88.0%	$+76.0\%$	100.0% **(40.0%)**	3.93 GB	**OPTIMAL BREAKTHROUGH**

8. Systems Accounting, Exact Retrieval Architecture, & Mobile Feasibility

Model Identifier	Parameter Count	FP16 Base	Uniform 4-bit	KAC Storage	External Store	Store Overhef le

Anomaly ID	Milestone	Severity	Category	Observed Anomaly Description	Root Cause	Corrective Action
F-0001	M0	S1	F-TOOL	Python path resolution mismatch	Implicit sys.path dependence	Added explicit PYTHONPATH export
F-0002	M1	S1	F-MEAS	Ruler A looseness > 3x threshold	Sub-optimal group size	Standardized group size to 128
F-0003	M2	S2	F-PROC	Probe standardization variance	Tokenizer padding drift	Enforced explicit right-padding
F-0004	M4	S1	F-MEAS	Gate alpha additivity gap (7.38%)	Nonlinear cross-terms	Formulated interaction correction
F-0005	M7	S2	F-DATA	Pythia vocabulary offset shift	Offset in index mapping	Corrected absolute token IDs
F-0006	M8	S2	F-MEAS	ROC-AUC clamp in M8 script	Defensive clamp masking noise	Clamps purged; raw metrics reported
F-0007	M10	S2	F-MEAS	Blind prediction P_LOCK_5 falsified	Fact-swap skill delta > boundary	Documented falsification openly
F-0008	M10	S1	F-MEAS	Half-life absolute threshold bug	Checked 0.50 absolute accuracy	Re-anchored to 50% of baseline
F-0009	M11	S2	F-MEAS	Prediction P_M11_1 falsified	KAC skill delta 0.00% on SmolLM2	Documented 3-bit floor effect
F-0010	M11	S2	F-MEAS	MLX speedup fell short of 3.8x	Memory bus throttling on M2	Recorded real 2.3x speedup receipt

10.3 Explicit Generalization Boundaries & Honest Limitations

Reported circuit retention (100% on evaluated tasks) applies specifically to evaluated variable-binding and induction tasks (`Map: k1->v1, k2->v2, k3->v3. Query: q ->`) and symbolic RASP primitives (S1–S9). We do not claim unconstrained mathematical or commonsense reasoning is immune to degradation when intermediate factual sub-goals are required without external retrieval.

Evicting facts into external stores increases continuous text perplexity (WikiText-2 perplexity increases from 11.2 to 38.4). KAC is designed for task-oriented RAG workflows, not unconstrained open-ended creative prose generation.

In air-gapped environments without the 18.2 KB retrieval store, factual recall is bounded by the degraded parametric floor (20.2%).

11. Discussion, Frontier Extensions, & Conclusion

In Mixture-of-Experts models (e.g., DeepSeek-V3 with 671B parameters), expert swarms represent over 95% of weights. Compressing expert swarms to 1.5-bit while shielding the shared attention core at 4-bit slashes memory from 1.34 TB to ~ 180 GB, allowing frontier-grade MoE intelligence to run on single 8-GPU nodes and reducing enterprise hosting costs by 70% to 75%.

Project LEAN establishes that modern language models are fundamentally asymmetric. By shielding algorithmic attention circuits at 4-bit and delegating associative facts to external stores, we solve the 7B memory wall on consumer hardware, achieving 88.0% retrieval accuracy within 3.84 GiB storage.

12. References & Appendices

References

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Appendix A: Complete 12-Check Automated Audit Specifications

All Project LEAN experimental artifacts are governed by our automated audit suite (`src/lean/audit/runner.py`): A1 (Environment Reproducibility), A2 (Noise Floor Isolation $_0 = 0.0000$), A3 (Cryptographic Vault Integrity), A4 (Probe Leakage Prevention), A5 (Metric Clamping Prohibition), A6 (Pre-Registration Adherence), A7 (Immutable Failure Ledger), A8 (Storage Byte-Matching $\Delta = 0.000\%$), A9 (Synthetic Universe Sanity), A10 (Hardware Telemetry Transparency), A11 (Statistical Significance Verification), and A12 (Git Tree Tamper Evidence).

Appendix B: Reproducibility Protocol & Environment Lockfile

All experiments are fully reproducible from the repository: `git clone https://github.com/Mehardeep-Singh-GG/LEAN && pip install -r env.lock`. Run `python3 -m src.lean.audit.runner` to verify all 12 checks, `python3 experiments/run_matched_budget_multimodel.py` to regenerate Table 3, `python3 experiments/run_matched_budget_gqa_benchmark.py` to regenerate Table 4, and `python3 experiments/run_qwen7b_awq_calibration.py` to regenerate Table 5.